



CubeSat Payload: Thermal Regulation Applications of E-Ink for Small Satellites

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Introduction

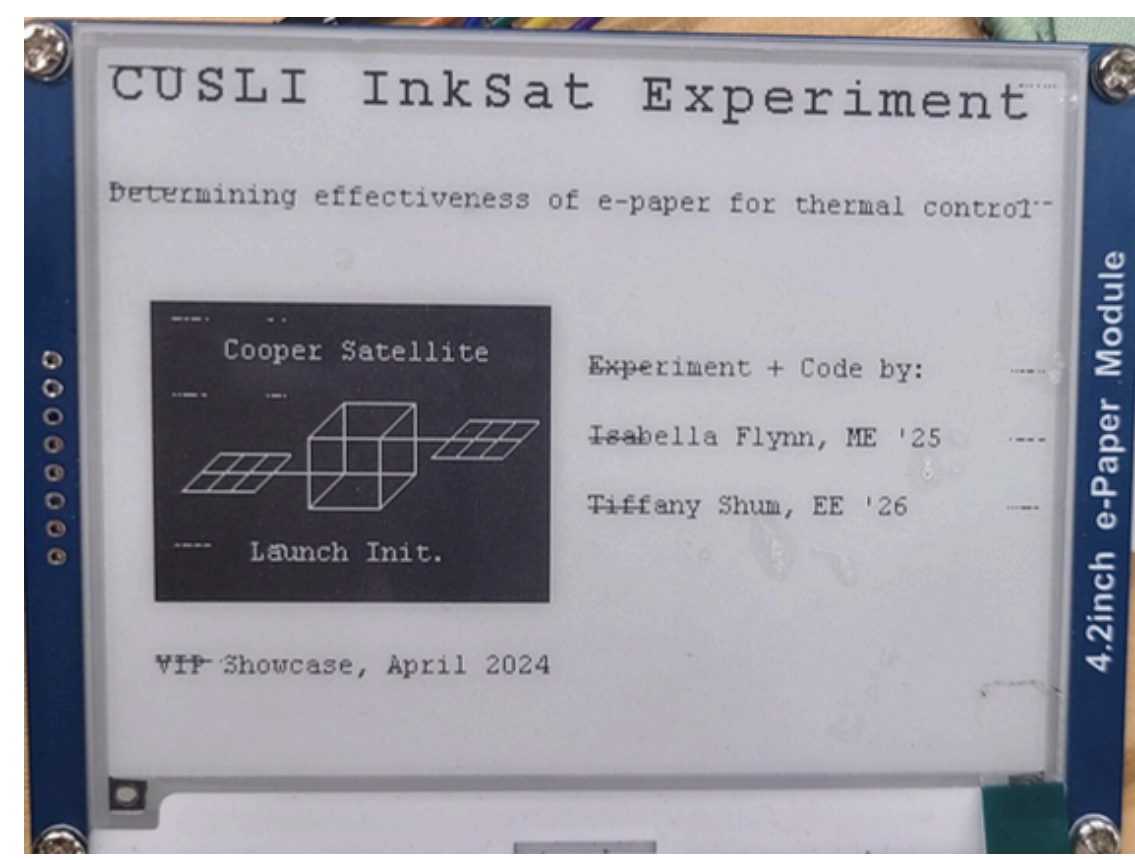
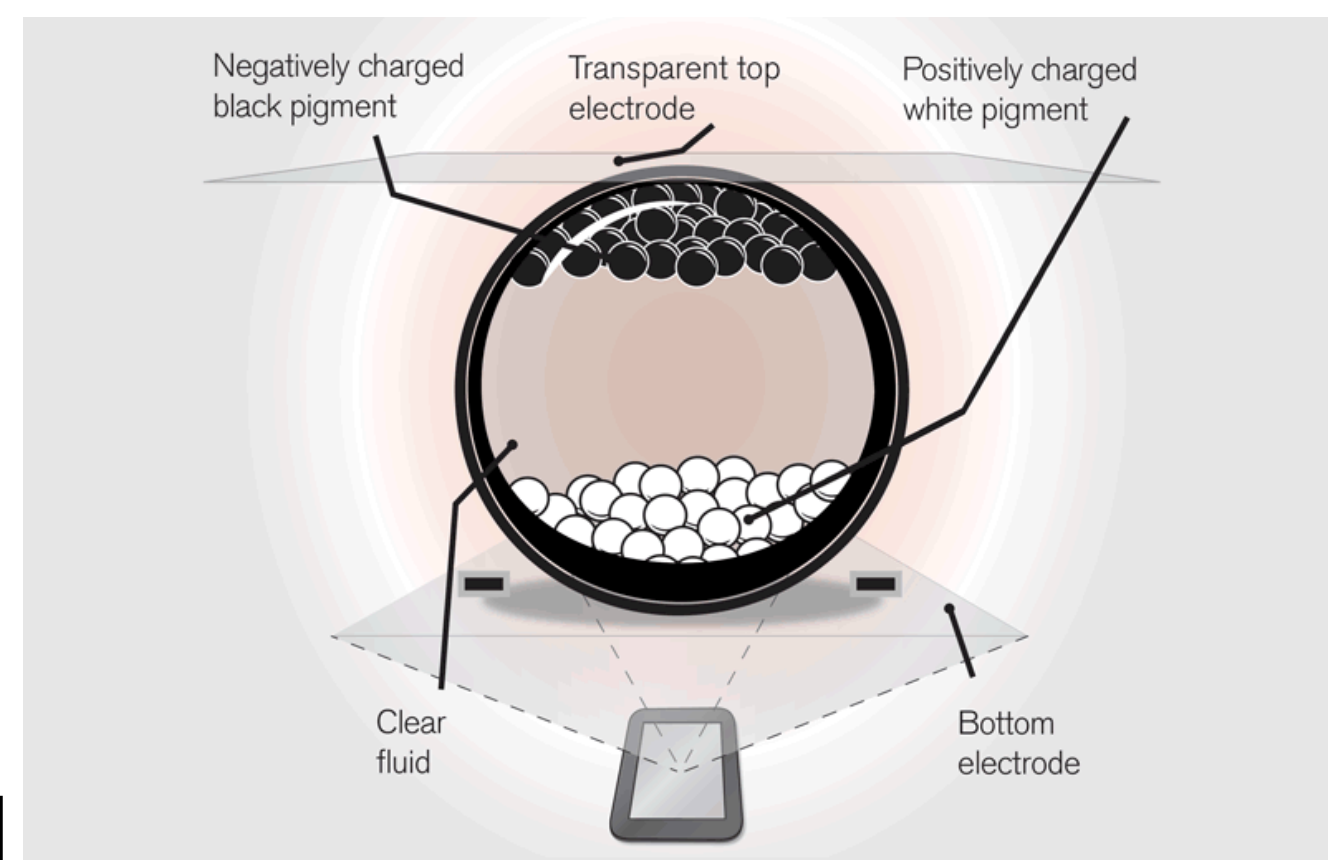
Satellites in low earth orbit (LEO) can go from an extremely cold environment (-65°C) to an extremely hot one (125°C) within the course of 4 hours [1].

For electronics that must operate at consistent temperatures and are currently heated only by other electronics, thermal regulation can be high consumers of energy.

This research explores a system consisting of a metal plate and an e-paper module at steady state, in which the thermal impact of radiation on the screen can be quantified using energy balances. The experiment was conducted several times in order to determine the emissivity of e-paper depending on its pixel density.

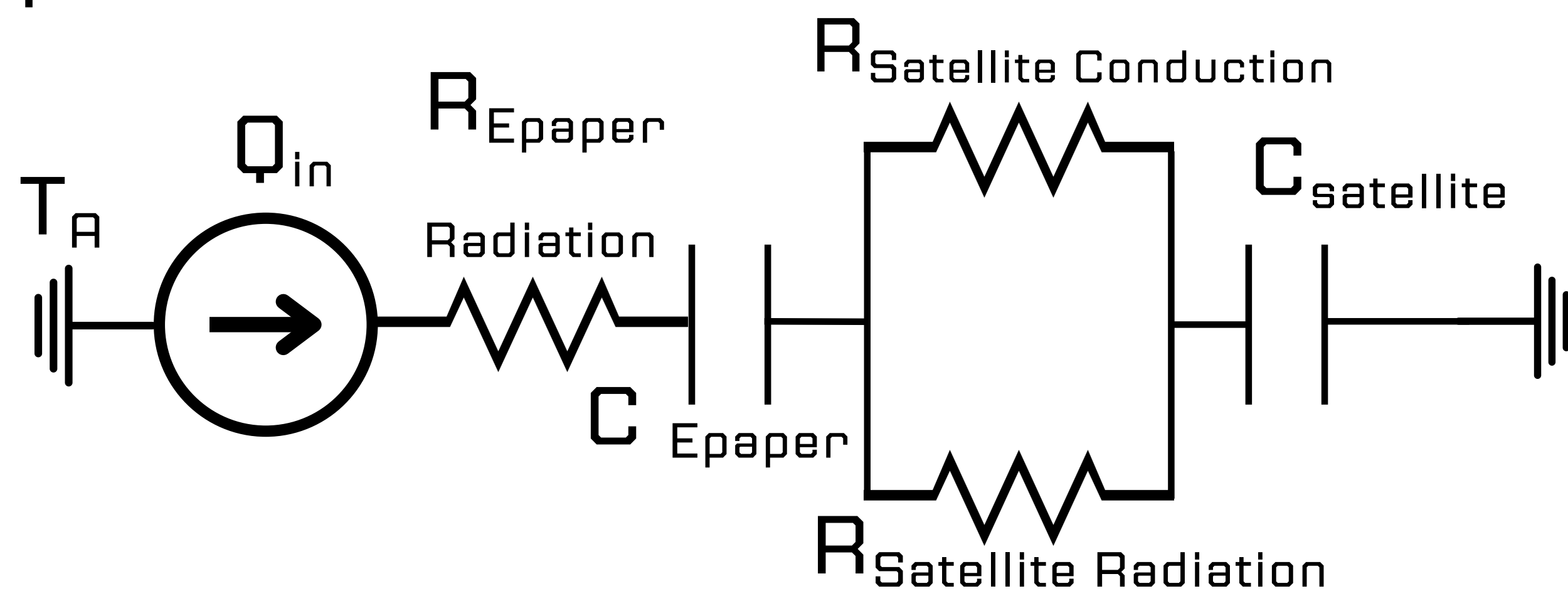
Understanding E-Paper

E-paper is an electrophoretic display that draws a near zero current when the display is not being changed. Because of this power efficiency setting, it is a compelling case study for this experiment.



Thermal Model

System Simplification: R indicates a thermal resistance, T indicates a temperature, Q indicates a heat flow, and C indicates a thermal capacitance.



The rate of heat flow within the psuedo-satellite is equal to that of the e-paper:

$$\frac{-kA_{surface}\Delta T}{\Delta z} = \epsilon\sigma(T_{surface}^4 - T_{source}^4)$$

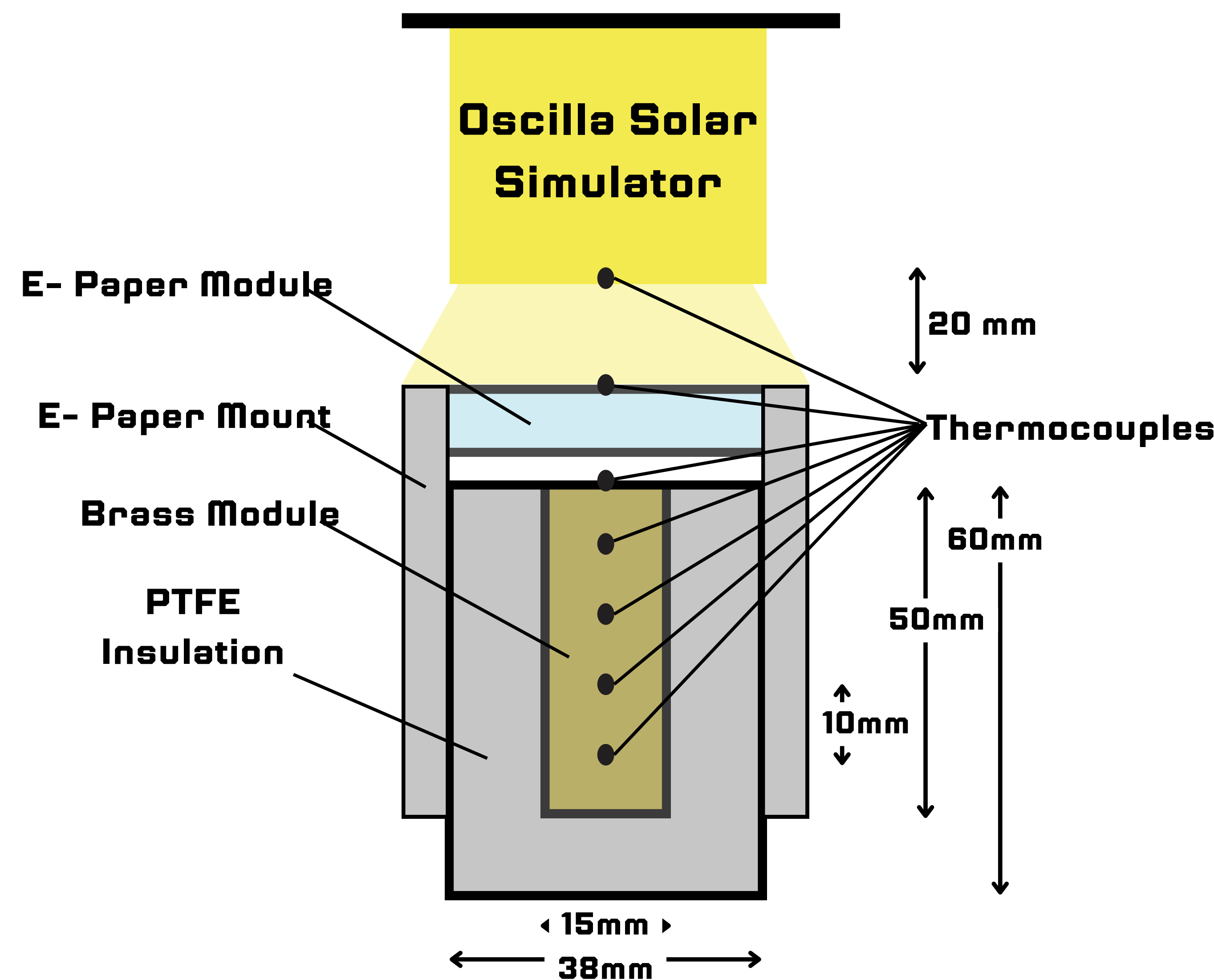
where ϵ is emissivity, σ is the Stefan Boltzman constant, Δz is the distance between the temperature measurements within the psuedo- satellite, and k is thermal conductivity.

Methodology and Setup

An Oscilla LED Solar Simulator is used to mimic sunlight in a controlled enironment. Within the beam produced by an 85mm distance from the bulb, and 15mm diameter, it prooduces a near perfect spectral match to the sun.

A brass-360 cylinder is used due to its high thermal conductivity which promotes heat transfer in the test module, and low cost.

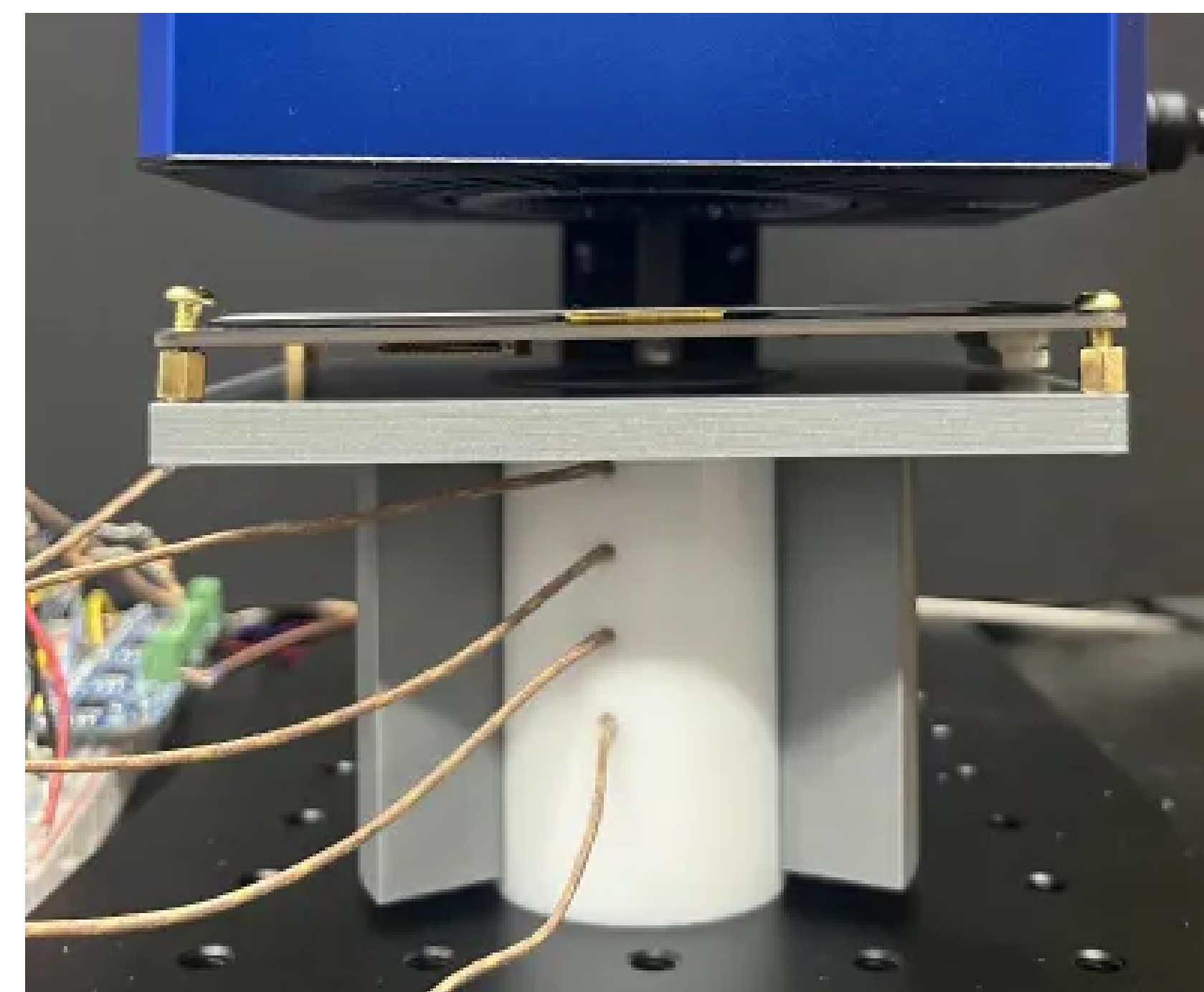
Type- K thermocouples are used to take both internal and surface temperatures, shown in the figure below, as they perform well in high temperature environemnts, including that of LED [3].



The system rests 20mm below the bulb of the solar simulator, to promote the rate of heat exchange while remaining in the constraints of the solar simulator's effective beam diameter.

The light is turned on and remains untouched until steady state is reached.

Steady state is determined by calculating the thermal conductivity of brass using the internal temperature measurements and Fourier's law of conductivity, and comparing that to the literature value for that of brass, 115 W/m-K.



Results

The experiment was conducted 3 times for the following cases: a control with brass only, a control with both brass and insulation, a case with a fully colored screen, and a case with a fully uncolored screen

The system without insulation was able to determine the thermal conductivity of brass within 74% error at steady state (appx 1.5 hours). However, the experiment was much more effective with an insulation layer, giving a thermal conductivity within 8% error at steady state (appx 1.25 hours). The following table shows the determined properties associated with each case.

Test	Experimental Thermal conductivity of brass (W/m-K)	Standard deviation	Literature thermal conductivity of brass (W/m-K)	% Error	Experimental emissivity of e- paper	Standard deviation
No insulation, no e-paper	29.82	± 5.31	115	74.07	N/A	N/A
Insulation, no e-paper	124.25	± 10.11		8.04	N/A	N/A
Insulation, fully uncolored e-paper	125.95	± 16.5		9.52	0.0143	± 0.00243
Insulation, fully colored e-paper	118.01	± 20.28		2.6	0.307	± 0.133

The system with a fully uncolored e-paper module was experimentally about 30x less emissive than the system with a fully colored e-paper. The ability of the fully colored e-paper to absorb heat was also notable during the tests, as the average surface temperature of the uncolored and fully colored e-paper were 44.13°C and 77.84°C , respectively.

However, there was significant variance between tests for the fully colored e-paper module. Moreover, there was significant variance in the prelliminary thermal conductivity calculation. This is likely due to the $\pm 2.2^{\circ}\text{C}$ error associated with the thermocouples. This may be reduced by switching to a type T thermocouple, which has an associated error of $\pm 1.0^{\circ}\text{C}$.

Acknowledgments

Thank you to the New York Space Grant Consortium for financial support of this research. Thank you to Dr. Rosen, Dr. Wright, and Dr. Sidebotham for helping in the development of both the model of the system and experimental methods. Thank you to Daniel Kim for being part of the initial brainstorming and methodology development process. Thank you to Tiffany Shum for helping interface with the e-paper. Thank you to Rina and Jay for support both in and out of the lab.

References

- [1] Plante, Jeannette, and Brandon Lee. "Environmental Conditions for Space Flight Hardware: A Survey." Ntrs.nasa.gov, 1 Jan. 2005, ntrs.nasa.gov/citations/20060013394.
- [2] "Electronic Paper Explained: What Is It and How Does It Work?" Visionect, 20 Feb. 2024, www.visionect.com/blog/electronic-paper-explained-what-is-it-and-how-does-it-work/.
- [3] Omega, "What is a thermocouple and how does it work?," Omega.co.uk, 2019. https://www.omega.co.uk/prodinfo/thermocouples.html